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METHOD OF MAKING A SEMICONDUCTOR DEVICE USING AN EDGE BEVEL REMOVAL SYSTEM CONTAINING A GAS NOZZLE

Abstract

A method of performing an edge bevel removal process includes providing a nozzle assembly including a liquid dispensation nozzle configured to dispense an etchant liquid and a gas dispensation nozzle configured to dispense a gas, and performing an etchant liquid dispensation process in which a stream of the etchant liquid is dispensed from an orifice of the liquid dispensation nozzle toward a peripheral region of a top surface of a device wafer while a stream of the gas is directed at a bottom surface of the liquid dispensation nozzle.

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Background/Summary

FIELD

[0001] The present disclosure relates generally to the field of semiconductor manufacturing, and specifically to a method of making a semiconductor device using an edge bevel removal system containing a gas nozzle and to the edge bevel removal system.

BACKGROUND

[0002] An edge bevel removal (“EBR”) process is employed in semiconductor device manufacturing to facilitate subsequent chemical mechanical polishing of electroplated copper on a wafer and to prevent or reduce delamination and/or contamination caused by residual copper at the wafer edge. The edge bevel removal process removes peripheral regions of the electroplated copper that may have an excessively high thickness and causes difficulties in a subsequently chemical mechanical polishing process, i.e., makes complete removal of the electroplated copper from the peripheral regions difficult. Edge bevel removal results in a thickness reduction or complete removal of copper near the edge of the wafer. For example, the edge bevel removal process removes a peripheral portion of the electroplated copper by spraying a wet etchant, i.e., an etchant liquid, from an EBR nozzle at the edge of the wafer covered in copper. However, the edge bevel removal process may remove too much copper or non-uniformly remove the copper at the wafer edge (i.e., at the bevel edge of the wafer).

SUMMARY

[0003] According to an aspect of the present disclosure, a method of performing an edge bevel removal process includes providing a nozzle assembly including a liquid dispensation nozzle configured to dispense an etchant liquid and a gas dispensation nozzle configured to dispense a gas, and performing an etchant liquid dispensation process in which a stream of the etchant liquid is dispensed from an orifice of the liquid dispensation nozzle toward a peripheral region of a top surface of a device wafer while a stream of the gas is directed at a bottom surface of the liquid dispensation nozzle.

[0004] According to another aspect of the present disclosure, an edge bevel removal apparatus comprises a chuck configured to hold a wafer thereupon, and a nozzle assembly comprising a liquid dispensation nozzle configured to dispense an etchant liquid and a gas dispensation nozzle configured to dispense a gas. The liquid dispensation nozzle and the gas dispensation nozzle extend substantially parallel to each other (e.g., having lengthwise axes within zero to five degrees of each other); the liquid dispensation nozzle includes a liquid orifice aimed at a peripheral portion of the chuck; the gas dispensation nozzle is located below and radially inward relative to the liquid dispensation nozzle; and the gas dispensation nozzle includes a gas orifice aimed at a bottom surface of the liquid dispensation nozzle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1A is a top-down view of a wafer on which an edge bevel removal process may be subsequently performed.

[0006] FIG. 1B is a vertical cross-sectional view of the wafer along the vertical plane B-B' of FIG. 1A.

[0007] FIG. 1C is a magnified view of region C of the wafer in FIG. 1B.

[0008] FIG. 2 illustrates a first mechanism for deleterious collateral etching of a conductive layer from within a region that should not be etched.

[0009] FIG. 3 illustrates a second mechanism for deleterious collateral etching of a conductive layer from within a region that should not be etched.

[0010] FIG. 4 illustrates operation of an edge bevel removal apparatus of the present disclosure during a normal operational mode according to an embodiment of the present disclosure.

[0011] FIG. 5 illustrates operation of the edge bevel removal apparatus of the present disclosure in case a wetting of a backside surface of a liquid dispensation nozzle occurs according to an embodiment of the present disclosure.

[0012] FIG. 6 illustrates operation of the edge bevel removal apparatus of the present disclosure in case droplets of an etchant liquid accidentally falls off the liquid dispensation nozzle according to an embodiment of the present disclosure.

[0013] FIG. 7 is a magnified view of a region of a wafer after performing the edge bevel removal process of the present disclosure.

DETAILED DESCRIPTION

[0014] As discussed above, the embodiments of the present disclosure are directed to an edge bevel removal system employing a gas nozzle which emits a gas stream (e.g., gas jet) below an EBR liquid etchant nozzle and a method of operating the same, the various aspects of which are described below.

[0015] The drawings are not drawn to scale. Multiple instances of an element may be duplicated where a single instance of the element is illustrated, unless absence of duplication of elements is expressly described or clearly indicated otherwise. Ordinals such as “first,” “second,” and “third” are employed merely to identify similar elements, and different ordinals may be employed across the specification and the claims of the instant disclosure. The term “at least one” element refers to all possibilities including the possibility of a single element and the possibility of multiple elements.

[0016] The same reference numerals refer to the same element or similar element. Unless otherwise indicated, elements having the same reference numerals are presumed to have the same composition and the same function. Unless otherwise indicated, a “contact” between elements refers to a direct contact between elements that provides an edge or a surface shared by the elements. If two or more elements are not in direct contact with each other or from each other, the two elements are “disjoined from” each other or “disjoined among” one another. As used herein, a first element located “on” a second element can be located on the exterior side of a surface of the second element or on the interior side of the second element. As used herein, a first element is located “directly on” a second element if there exist a physical contact between a surface of the first element and a surface of the second element. As used herein, a first element is “electrically connected to” a second element if there exists a conductive path consisting of at least one conductive material between the first element and the second element. As used herein, a “prototype” structure or an “in-process” structure refers to a transient structure that is subsequently modified in the shape or composition of at least one component therein.

[0017] The edge bevel removal system containing a gas nozzle below the EBR wet etchant nozzle of the embodiments present disclosure prevents or reduces unintentional dripping of an etchant liquid from the EBR wet etchant nozzle within an area of a metal that is not removed by the edge bevel removal process, thereby preventing or reducing undesirable overetching of the metal from the wafer bevel edge. The gas nozzle directs a gas steam (e.g., gas jet) toward a backside surface of the EBR wet etchant nozzle. A stream (e.g., jet) of the etchant liquid is dispensed from an orifice of the nozzle toward a peripheral (i.e., edge) region of a top surface of a wafer during the edge bevel removal process, while the gas steam is directed at a bottom surface of the nozzle such that the gas stream flows along the bottom surface of the nozzle toward the orifice of nozzle. The gas steam prevents or reduces unintended dripping of droplets of the etchant liquid from the EBR wet etchant nozzle inside the inner edge of the annular edge bevel removal area. Therefore, the process yield of subsequently chemical mechanical polishing process can be increased. The various aspects of the present disclosure are now described with reference to accompanying drawings.

[0018] FIGS. 1A-1C illustrate a wafer that is processed with an edge bevel removal process known in the art. A device wafer **10** can include a substrate **2**, such as semiconductor substrate (e.g., a silicon wafer), a device layer **4** including semiconductor devices (such as a logic circuit including

field effect transistors or a memory array including a two-dimensional array of memory elements) and metal interconnect structures embedded within dielectric material layers, and a metallic (i.e., electrically conductive) layer (6, 7, 8) overlying the device layer 4. The metallic layer may be subsequently patterned to form various additional metal interconnect structures or metal pads. The metallic layer (6, 7, 8) is magnified along the vertical direction with a higher magnification scale than the device layer 4 and the semiconductor substrate 2 to illustrate components thereof. The device layer 4 includes various devices and patterns that are repeated with a two-dimensional periodicity to define a two-dimensional array of semiconductor dies 9 on the device wafer 10. After completion of all subsequent processing steps on the device wafer 10, the device wafer 10 may be diced along dicing channels between neighboring pairs of semiconductor dies 9 to singulate the semiconductor dies 9.

[0019] The metallic layer (6, 7, 8) may comprise a metallic barrier liner 6, a copper seed layer 7, and an electroplated copper layer 8. The metallic barrier liner 6 typically comprises a conductive metallic nitride, such as TiN, TaN, WN, and/or MoN, and/or a refractory metal, such as Ti or Ta, may be deposited by physical vapor deposition or chemical vapor deposition, and may have a thickness in a range from 5 nm to 100 nm, although lesser and greater thicknesses may also be employed. The copper seed layer 7 consists essentially of copper, may be deposited by physical vapor deposition, and may have a thickness in a range from 20 nm to 100 nm, although lesser and greater thicknesses may also be employed. The electroplated copper layer 8 consists essentially of copper, may be deposited by electroplating, and may have a thickness in a range from 500 nm to 10,000 nm, although lesser and greater thicknesses may be employed. The electroplated copper layer 8 may be deposited employing a lip seal that covers a peripheral annular area of the device wafer 10 during the electroplating process that forms the electroplated copper layer 8. An “edge cut” feature is present in the electroplated copper layer 8, which means that the electroplated copper layer 8 is not deposited within an edge exclusion distance d_{ee} from the outermost periphery of the device wafer 10. The edge exclusion distance d_{ee} is typically about 1 mm, although lesser and greater dimensions may also be employed. In alternative embodiments, the metallic barrier liner 6 and/or the copper seed layer 7, may be omitted. In other alternative embodiments, a metal other than copper layer 8 may be present and removed by the ERB process.

[0020] An edge bevel removal process can be subsequently performed to remove portions of the copper seed layer 7 and the electroplated copper layer 8 that are located within a uniform distance from the outermost periphery of the device wafer 10. This uniform distance is herein referred to as an edge bevel removal distance d_{ebr} . The edge bevel removal process completely removes the portions of the copper seed layer 7 and the electroplated copper layer 8 that are located in a “bevel region” of the device wafer 10, which refers to the portion of the device wafer 10 in which the semiconductor substrate 2 has a beveled top surface, i.e., a non-horizontal top surface. Further, the edge bevel removal distance d_{ebr} can be set such that additional portions of the copper seed layer 7 and the electroplated copper layer 8 that are located over a horizontally-extending portion of the top surface of the semiconductor substrate 2 and proximal to the bevel region of the wafer are also removed. Typically, the edge bevel removal distance d_{ebr} can be set in a range from 1.5 mm to 3 mm, although lesser and greater edge bevel removal distances d_{ebr} may also be employed.

[0021] The edge bevel removal process can be performed by mounting the device wafer 10 on a chuck, and by directing an etchant liquid that etches copper toward a peripheral portion of the layer stack including the copper seed layer 7 and the electroplated copper layer 8. The etchant liquid 80 may comprise any solution that etches copper. Typically, the etchant liquid 80 is selected to etch copper selective to the material of the metallic barrier liner 6 to prevent collateral etching of the materials in the device layer 4. For example, the etchant liquid 80 may comprise a mixture of sulfuric acid and hydrogen peroxide. The etchant liquid 80 is applied radially outward while the device wafer 10 is spinning on a rotating chuck to ensure that the etchant liquid 80 stays outside an area defined by the lateral inward offset by the edge bevel removal distance d_{ebr} from the outmost

periphery of the device wafer **10**. If a notch or a flat is present on the device wafer **10**, the boundary for keeping the etchant liquid **80** outward from may be suitably adjusted near the notch or the flat. [0022] During application of the etchant liquid to the periphery of the device wafer **10**, a dispensation nozzle (i.e., the EBR wet etchant nozzle) for applying the etchant liquid is positioned over the device wafer **10** and is oriented radially outward such that the movement direction of the dispensed etchant liquid is outward. Further, the spinning of the device wafer **10** applies a centrifugal force to the dispensed etchant liquid. The combination of the dispensation direction of the etchant liquid and the centrifugal force tends to keep the dispensed etchant liquid outside a protected area of the device wafer **10** that is defined by the lateral inward offset by the edge bevel removal distance d_{ebr} from the outmost periphery of the device wafer **10**.

[0023] However, as shown in FIG. **1C**, the present inventors realized that collaterally etched regions **89** within the protected area of the device wafer **10** may be unintentionally etched by the etchant liquid. The thickness of the electroplated copper layer **8** has a local reduction in the collaterally etched regions **89**. This feature has a deleterious effect on a subsequent chemical mechanical polishing step in which severe dishing of the underlying portion of the top surface of the device layer **4** can occur, and structures proximate to the underlying portion of the top surface of the device layer **4** can be damaged. The distribution of the collaterally etched regions **89** may be random, and is believed to be due to the operational principle of an edge bevel removal apparatus employed to perform the edge bevel removal process.

[0024] FIG. **2** illustrates a first mechanism for a deleterious collateral etching of a material layer during the EBR process from within a region that should not be etched. During the edge bevel removal process illustrated in FIG. **2**, the liquid dispensation nozzle (i.e., the EBR wet etchant nozzle) **30** is positioned over the device wafer **10**, and is directed downward and radially outward toward a peripheral region of device wafer **10**. During dispensation of an etchant liquid **80** from the orifice of the liquid dispensation nozzle **30**, the etchant liquid **80** (e.g., mixture of sulfuric acid and hydrogen peroxide for etching copper) is directed at the peripheral area of the device wafer **10**. The centrifugal force generated by the spinning of the device wafer **10** and applied to the portion of the etchant liquid **80** on the device wafer **10** causes the portion of the etchant liquid **80** on the device wafer **10** to stay farther outward than a threshold distance from the center of the device wafer **10**. The minimum distance between the portion of the etchant liquid **80** on the device wafer **10** and the center of the device wafer **10** defines the protection region of the device wafer **10**. The applied etchant liquid **80** remains in the edge bevel removal area (i.e., the EBR area), which is the area located outside the protected area.

[0025] The etchant liquid **80** is subjected to various forces while being dispensed out of the orifice of the liquid dispensation nozzle **30**. A phenomenon known as Coanda effect occurs at the bottom of the orifice of the liquid dispensation nozzle **30** and on the bottom surface of the liquid dispensation nozzle **30**. The Coanda effect is a phenomenon in which a fluid adheres to and follows the contour of a surface near an orifice out of which a main flow of the liquid is discharged. The Coanda effect causes a portion of a liquid to stay attached to a surface, and especially a convex surface, which is proximate to the orifice of the liquid dispensation nozzle **30**.

[0026] The Coanda effect around the orifice of the liquid dispensation nozzle **30** causes a backside droplet **8B** of the etchant liquid **80** to accumulate and to grow on a bottom surface segment and/or a backside surface segment of the liquid dispensation nozzle **30**. When the backside droplet **8B** accumulates a sufficient volume, the force of gravity applied to the backside droplet **8B** can exceed the surface tension between the backside droplet **8B** and the backside surface of the liquid dispensation nozzle **30**, and a portion of the backside droplet **8B** falls onto a protection region of the wafer as a surface droplet **8S** of the etchant liquid **80**. The surface droplet **8S** is rapidly detached from the remaining portion of the backside droplet **8B** because the liquid dispensation nozzle **30** is stationary and the device wafer **10** is rotating. However, the surface droplet **8S** can etch a significant portion of the metallic layer underneath, which includes a stack of a copper seed

layer **7** and an electroplated copper layer **8**. A collaterally etched region **89** described with reference to FIGS. **1A-1C** may be formed in this manner. This mode of formation of collaterally etched regions **89** is herein referred to as a Coanda collateral etch mode.

[0027] FIG. **3** illustrates a second mechanism for a deleterious collateral etching of a material layer from within a region that should not be etched. In this case, the etchant liquid **80** may remain outside the protection region during the steady state of the edge bevel removal process in which the device wafer **10** rotates, and the dispensed portion of the etchant liquid **80** is continually pushed outward due to the centrifugal force. Even through the orifice of the liquid dispensation nozzle **30** is located within the area of the protection region in a top-down view, this arrangement does not cause a problem because a stream (e.g., jet) of the etchant liquid **80** is dispensed with a sufficient velocity to avoid falling of the etchant liquid **80** in an area that underlies the liquid dispensation nozzle **30**. Upon termination of the liquid dispensation step, however, a residual drop **8R** of the etchant liquid **80** may be present at the orifice of the liquid dispensation nozzle **30**. When the residual drop **8R** of the etchant liquid **80** falls onto the wafer, the residual drop **8R** does not have any lateral velocity, and falls within the protection region of the device wafer **10** that should not be etched by the etchant liquid **80**. A collaterally etched region **89** described with reference to FIGS. **1A-1C** may be formed in this manner. This mode of formation of collaterally etched regions **89** is herein referred to as a residual solution drop mode, or a residual etchant drop mode.

[0028] Additionally or alternatively, the flow velocity of the jet of the etchant liquid **80** can be poorly defined during initiation of the application of the jet of the etchant liquid **80** onto the device wafer **10**. The dispensed portion of the etchant liquid **80** from the orifice of the liquid dispensation nozzle **30** during turn-on of the stream of the etchant liquid **80** may have insufficient dispensation speed or excessive dispensation speed due to the transient variations in the pressure at the orifice of the liquid dispensation nozzle **30**, and spillage of the etchant liquid **80** within the protected area of device wafer **10** may occur, resulting in formation of collaterally etched regions **89**.

[0029] The size of the collaterally etched regions **89** may be reduced if a diluted etchant providing a lower etch rate is employed for the edge bevel removal process. However, use of such a diluted etchant results in a longer etch time, and inefficient utilization of the edge bevel removal tools.

[0030] According to an aspect of the present disclosure, an edge bevel removal apparatus includes a gas dispensation nozzle **40** which eliminates or reduces formation of collaterally etched regions **89** during or after the edge bevel removal process. The embodiment edge bevel removal apparatus comprises: a chuck **20** configured to hold a device wafer **10** thereupon; a nozzle assembly **100** comprising a liquid dispensation nozzle **30** configured to dispense an etchant liquid **80**, and a gas dispensation nozzle **40** configured to dispense a gas (e.g., a gas stream) **90**; and a process controller **500** configured to control the liquid dispensation nozzle **30** and the gas dispensation nozzle **40** and configured to perform an etchant liquid dispensation process in which a stream (e.g., jet) of the etchant liquid **80** is dispensed from an orifice **31** of the liquid dispensation nozzle **30** toward a peripheral region of a top surface of a device wafer **10** while a stream (e.g., jet) of the gas **90** is directed at a bottom surface of the liquid dispensation nozzle **30** such that a gas stream (e.g., jet) **90** flows along the bottom surface of the liquid dispensation nozzle **30** toward the orifice **31** of the liquid dispensation nozzle **30**. The top surface of the device wafer **10** may be the top surface of the metallic layer (**6**, **7**, **8**), which may include a metallic barrier liner **6**, a copper seed layer **7**, and an electroplated copper layer **8**.

[0031] The etchant liquid **80** comprises a liquid phase etchant that etches the material of the metallic layer (**6**, **7**, **8**). In one embodiment, the etchant liquid **80** comprises a liquid phase etchant that etches copper in the stack of the copper seed layer **7** and the electroplated copper layer **8** selective to the metallic nitride material in the metallic barrier liner **6**. In one embodiment, the etchant liquid **80** may comprise a mix of sulfuric acid and hydrogen peroxide. The gas **90** comprises an inert gas such as dry air, nitrogen, and/or argon.

[0032] The etchant liquid dispensation process is initiated while the device wafer **10** is mounted on

a chuck **20**, and rotates around an axis passing through the geometrical center of the device wafer **10** and through the geometrical center of the chuck **20**. Generally, the chuck **20** may have a shape of a circular plate, and the device wafer **10** can be mounted on the top surface of the chuck **20** such that the geometrical center of the device wafer **10** is positioned at a vertical axis passing through the geometrical center of the chuck **20**. The device wafer **10** can be attached to the chuck **20** through vacuum suction and/or through side chucks. In one embodiment, the chuck **20** may comprise three or more support points with side chucks (e.g., side barriers or restraints) which mechanically secure the device wafer **10** on the chuck **20** support points. A chuck rotation mechanism **18** can be provided underneath the chuck **20** to rotate the chuck **20** and the device wafer **10** around the vertical axis passing through the center of the chuck **20**. The angular speed of rotation of the device wafer **10** and the chuck **20** during the etchant liquid dispensation process is selected to ensure that sufficient centrifugal force is applied to a dispensed portion of the etchant liquid **80**, and may be in a range from **150** revolutions per minute (rpm) to 400 rpm, such as from 225 rpm to 350 rpm, although lesser and greater angular speeds of rotation may also be employed. [0033] The gas **90** stream may have laminar gas flow characteristics. A laminar flow refers to a smooth, orderly movement of a fluid in which adjacent layers of the fluid glide parallel among one another without significant mixing or disruption. In a laminar flow, the fluid particles follow well-defined paths, and the flow pattern is characterized by a lack of significant turbulence. According to an aspect of the present disclosure, the gas dispensation nozzle **40** is positioned such that a terminal portion of the laminar flow of the gas **90** stream transfers momentum to the stream of the etchant liquid **80** along a lateral direction toward an edge of the top surface of the device wafer **10**. In one embodiment, the stream of the etchant liquid **80** is directed toward the top surface of the device wafer **10** at a first non-zero and non-orthogonal tilt angle $\alpha 1$ relative to a horizontal plane including the top surface of the device wafer **10**. In one embodiment, the stream of the gas **90** is directed toward the top surface of the device wafer **10** at a second non-zero and non-orthogonal tilt angle $\alpha 2$ relative to the horizontal plane including the top surface of the device wafer **10**. In this case, the jet of the gas **90** transfers downward and outward momentum to a portion of the etchant liquid **80** that is present on the top surface of the device wafer **10**, and thereby assists in outward movement of the dispensed portion of the etchant liquid **80** on the top surface of the device wafer **10**.

[0034] The gas dispensation nozzle **40** can be located on the backside of the liquid dispensation nozzle **30**. In one embodiment, the orifice **41** of the gas dispensation nozzle **40** may be offset inward along a radial direction of the device wafer **10** from the backside of the liquid dispensation nozzle **30**. In one embodiment, the axial direction and the dispensation direction of the gas dispensation nozzle **40** may be parallel to the axial direction and the dispensation direction of the liquid dispensation nozzle **30**. In this case, the second non-zero and non-orthogonal tilt angle $\alpha 2$ may be the same as the first non-zero and non-orthogonal tilt angle $\alpha 1$. Alternatively, second non-zero and non-orthogonal tilt angle $\alpha 2$ may be smaller than the first non-zero and non-orthogonal tilt angle $\alpha 1$ by a differential angle ($\alpha 1 - \alpha 2$), which may be in a range from 0.01 degree to 5 degrees. The value of the first non-zero and non-orthogonal tilt angle $\alpha 1$ may be in a range from 40 degrees to 55 degrees, such as 45 degrees to 50 degrees although lesser and greater values may also be employed.

[0035] In one embodiment, the stream (e.g., jet) of the gas **90** is directed at a point within the top surface of the device wafer **10** that is more distal from an edge of the top surface of the device wafer **10** than a most distal surface of a portion of the etchant liquid **80** that is present on the top surface of the device wafer **10**. This alignment ensures that the momentum transfer from the stream of the gas **90** to the dispensed portion of the etchant liquid **80** is always radially outward, and splashing of the etchant liquid **80** by the stream of the gas **90** does not occur or is minimized.

[0036] In one embodiment, the liquid dispensation nozzle **30** comprises a conical frustum portion **30F** containing the orifice **31** at a narrow end thereof and further comprises a cylindrical portion

30C that is adjointed to a rear of the conical frustum portion **30F**. The jet of the gas **90** is directed at a bottom surface of the cylindrical portion **30C**. The laminar flow of the gas **90** extends underneath the cylindrical portion **30C** and the conical frustum portion **30F**. In one embodiment, the nozzle assembly **100** comprises an optional clamp structure **50** configured to hold the gas dispensation nozzle **40** at a fixed position relative to a position of the liquid dispensation nozzle **30**. In one embodiment, the liquid dispensation nozzle **30** is oriented such that a primary spray direction of the stream of the gas **90** is parallel to a primary spray direction of the stream of etchant liquid **80** from the liquid dispensation nozzle **30**.

[0037] In one embodiment, the nozzle assembly **100** comprises a first conduit **32** affixed to a rear of the liquid dispensation nozzle **30**, attached to the clamp structure **50**, and configured to flow the etchant liquid **80** therethrough, and a second conduit **42** affixed to a rear of the gas dispensation nozzle **40**, attached to the clamp structure **50**, and configured to flow the gas **90** therethrough. In one embodiment, the first conduit **32** has sufficient mechanical rigidity and strength for mounting into a first opening in the clamp structure **50**, and a second conduit **42** has sufficient mechanical strength for mounting to a second opening in the clamp structure **50**. The outlet of the first conduit **32** can be connected to a supply-side opening of the liquid dispensation nozzle **30**, and the outlet of the second conduit **42** can be connected to a supply-side opening of the gas dispensation nozzle **40**.

[0038] The first conduit **32** can be connected to a reservoir of the etchant liquid (i.e., a liquid etchant supply) through a first liquid supply pipe **34**, a first mass flow controller (e.g., valve) **36** configured to control the flow rate of the etchant liquid **80** therethrough, and a second liquid supply pipe **38**. The second conduit **42** can be connected to a gas supply (which may be a pressurized gas (e.g., nitrogen or argon) tank or an in-house gas (e.g., air) supply system) through a first gas supply pipe **44**, a second mass flow controller (e.g., valve) **46** configured to control the flow rate of the gas **90** therethrough, and a second gas supply pipe **48**. The process controller **500** can set the target values for the flow rate of the etchant liquid **80** through the first mass flow controller **36**. Further, the process controller **500** can set the target values for the flow rate of the gas **90** through the second mass flow controller **46**. Thus, the process controller **500** can control the flow rate of the etchant liquid **80** out of the liquid dispensation nozzle **30**, and can control the flow rate of the gas **90** out of the gas dispensation nozzle **40**.

[0039] In one embodiment, the edge bevel removal apparatus may also comprise an optional dispenser actuation mechanism **60** and a dispenser actuation arm **62** that are configured to laterally and/or vertically move and/or rotate the nozzle assembly **100**. In one embodiment, the dispenser actuation arm **62** may be connected to the clamp structure **50**. In one embodiment, an optional video camera **70** may be provided for monitoring the location of the edge of the dispensed portion of the etchant liquid **80** on the top surface of the device wafer **10**. Alternatively, the video camera **70** may be omitted.

[0040] Generally, a stream of the etchant liquid **80** is dispensed from an orifice **31** of the liquid dispensation nozzle **30** toward a peripheral region of a top surface of a device wafer **10** while a stream of the gas **90** is directed at a bottom surface of the liquid dispensation nozzle **30** during the etchant liquid dispensation process. A laminar flow of the gas **90** flows along the bottom surface of the liquid dispensation nozzle **30** toward the orifice **31** of the liquid dispensation nozzle **30** during the etchant liquid dispensation process.

[0041] In one embodiment, the process controller **500** may be programmed to perform a pre-dispensation preparation process in which the stream of gas **90** is turned on while the etchant liquid **80** is not dispensed from the orifice **31** of the liquid dispensation nozzle **30** prior to performing the etchant liquid dispensation process. This mode of operation prevents or reduces formation of collaterally etched regions **89** by causing any droplet of the etchant liquid **80** that may form around the orifice **31** of the liquid dispensation nozzle **30** prior to performing the etchant liquid dispensation process to fly off outward away from the protection region of the device wafer **10** that should be protected from the etchant liquid **80**.

[0042] In one embodiment, the process controller **500** may be programmed to perform a post-dispensation purge process in which the stream of gas **90** is turned on while the etchant liquid **80** is not dispensed from the orifice **31** of the liquid dispensation nozzle **30** after performing the etchant liquid dispensation process. This mode of operation prevents or reduces formation of collaterally etched regions **89** by causing any droplet of the etchant liquid **80** that may form around the orifice **31** of the liquid dispensation nozzle **30** after performing the etchant liquid dispensation process to fly off outward away from the protection region of the device wafer **10** that should be protected from the etchant liquid **80**.

[0043] In one embodiment, the process controller **500** may be programmed to perform a series of steps during the etchant liquid dispensation process. For example, the etchant liquid dispensation process may comprise: a main dispensation step during which the stream of gas **90** is turned on at a normal flow rate; a commencement step during which the stream of gas **90** is turned on at a first enhanced rate that is higher than the normal flow rate; and a termination step during which the stream of gas **90** is turned on at a second enhanced rate that is higher than the normal flow rate.

[0044] In one embodiment, the process controller **500** may be programmed to monitor a periphery of a dispensed portion of the etchant liquid **80** over the top surface of the device wafer **10** employing the optional video camera **70**, and to maintain a distance between and edge of the top surface of the device wafer **10** and the periphery of the dispensed portion of the etchant liquid **80** within a predetermined distance range, which may be the target value $\pm$ maximum deviation tolerance for the edge bevel removal distance d_{ebr} .

[0045] In one embodiment, the device wafer **10** comprises a semiconductor wafer **2** including semiconductor devices and metal interconnect structures (e.g., device layer **4**) thereupon, a continuous copper layer (**7, 8**) is present on the top surface of the device wafer **10**, and the etchant liquid **80** etches copper.

[0046] The embodiment edge bevel removal apparatus avoids or reduces the deleterious effects of the Coanda effect as discussed with reference to FIG. **2** and/or the unintended dripping of droplets of the etchant liquid **80** as discussed with reference to FIG. **3**.

[0047] Referring to FIG. **5**, the mechanism for suppressing growth of a backside droplet **8B** on the bottom side of the liquid dispensation nozzle **30** is illustrated. The stream of the gas **90** provides a laminar flow beneath the backside of the liquid dispensation nozzle **30**, and constantly transfers lateral momentum to any backside droplet **8B** that may begin to form adjacent to the orifice **31** of the liquid dispensation nozzle **30**. The constant transfer of the momentum from the stream of the gas **90** to the backside droplet **8B** prevents or reduces growth of the backside droplet **8B** to a sufficient size for the gravitational force on the backside droplet **8B** to overcome the surface tension of the backside droplet **8B**. Thus, even if a backside droplet **8B** were to be formed underneath the orifice **31** of the liquid dispensation nozzle **30**, the backside droplet **8B** does not fall on the top surface of the device wafer **10**.

[0048] Referring to FIG. **6**, the mechanism for preventing or reducing dropping of droplet **8D** of the etchant liquid **80** on the bottom side of the liquid dispensation nozzle **30** is illustrated. The stream of the gas **90** provides a laminar flow beneath the backside of the liquid dispensation nozzle **30** prior to and after the etchant liquid dispensation process. The droplet **8D** of the etchant liquid **80** that may form around the orifice **31** of the liquid dispensation nozzle **30** is blown away from above the protection region of the device wafer **10** into the area of the edge bevel removal region that is located within the edge bevel removal distance d_{ebr} from the outermost periphery of the device wafer **10**. Thus, the droplet **8D** of the etchant liquid **80**, if present, falls into the area of the edge bevel removal region, and does not fall on the top surface of the device wafer **10** in the protection region.

[0049] Referring to FIG. **7**, is a magnified view of a region of a device wafer **10** is illustrated after performing the embodiment edge bevel removal process. The device wafer **10** after the embodiment edge bevel removal process may be free of any collaterally etched region **89** or may

include a lower number of the collaterally etched regions **89**. Thus, the copper layer (**7, 8**) on the top surface of the device wafer **10** has a more uniform thickness distribution, and the process yield of a subsequent chemical mechanical polishing process can be enhanced. In one embodiment, the edge bevel removal distance d_{ebr} may be reduced to a range from 1.2 mm to 2.0 mm due to the enhanced control of the distribution area of the etchant liquid **80**. A larger fraction of the entire area of the device wafer **10** may be employed for manufacture of devices through use of the embodiment edge bevel removal apparatus.

[0050] Although the foregoing refers to particular preferred embodiments, it will be understood that the disclosure is not so limited. It will occur to those of ordinary skill in the art that various modifications may be made to the disclosed embodiments and that such modifications are intended to be within the scope of the disclosure. Compatibility is presumed among all embodiments that are not alternatives of one another. The word “comprise” or “include” contemplates all embodiments in which the word “consist essentially of” or the word “consists of” replaces the word “comprise” or “include,” unless explicitly stated otherwise. Whenever two or more elements are listed as alternatives in a same paragraph or in different paragraphs, a Markush group including a listing of the two or more elements is also impliedly disclosed. Whenever the auxiliary verb “can” is employed in this disclosure to describe formation of an element or performance of a processing step, an embodiment in which such an element or such a processing step is not performed is also expressly contemplated, provided that the resulting apparatus or device can provide an equivalent result. As such, the auxiliary verb “can” as applied to formation of an element or performance of a processing step should also be interpreted as “may” or as “may, or may not” whenever omission of formation of such an element or such a processing step is capable of providing the same result or equivalent results, the equivalent results including somewhat superior results and somewhat inferior results. Where an embodiment employing a particular structure and/or configuration is illustrated in the present disclosure, it is understood that the present disclosure may be practiced with any other compatible structures and/or configurations that are functionally equivalent provided that such substitutions are not explicitly forbidden or otherwise known to be impossible to one of ordinary skill in the art. If publications, patent applications, and/or patents are cited herein, each of such documents is incorporated herein by reference in their entirety.

Claims

1. A method of performing an edge bevel removal process, comprising: providing a nozzle assembly comprising a liquid dispensation nozzle configured to dispense an etchant liquid and a gas dispensation nozzle configured to dispense a gas, wherein the gas dispensation nozzle is located below the liquid dispensation nozzle, and is radially offset inward relative to the liquid dispensation nozzle; and performing an etchant liquid dispensation process in which a stream of the etchant liquid is dispensed from an orifice of the liquid dispensation nozzle toward a peripheral region of a top surface of a device wafer while a stream of the gas is directed at a bottom surface of the liquid dispensation nozzle such that a laminar flow of the gas flows along the bottom surface of the liquid dispensation nozzle toward the orifice of the liquid dispensation nozzle.
2. The method of claim 1, wherein a terminal portion of the laminar flow of the gas transfers momentum to the stream of the etchant liquid along a lateral direction toward an edge of the top surface of the device wafer.
3. The method of claim 1, wherein the stream of the gas is directed toward the top surface of the device wafer at a non-zero and non-orthogonal tilt angle relative to a horizontal plane including the top surface of the device wafer such that the stream of the gas transfers downward and outward momentum to a portion of the etchant liquid that is present on the top surface of the device wafer.
4. The method of claim 1, wherein the stream of the gas is directed at a point within the top surface of the device wafer that is more distal from an edge of the top surface of the device wafer than a

most distal surface of a portion of the etchant liquid that is present on the top surface of the device wafer.

5. The method of claim 2, wherein: the liquid dispensation nozzle comprises a conical frustum portion containing the orifice at a narrow end thereof and further comprises a cylindrical portion that is adjoined to a rear of the conical frustum portion; the stream of the gas is directed at a bottom surface of the cylindrical portion; and the laminar flow of the gas extends underneath the cylindrical portion and the conical frustum portion.

6. The method of claim 1, wherein the nozzle assembly further comprises a clamp structure configured to hold the gas dispensation nozzle at a fixed position relative to a position of the liquid dispensation nozzle.

7. The method of claim 6, wherein the nozzle assembly further comprises: a first conduit affixed to a rear of the liquid dispensation nozzle, attached to the clamp structure, and configured to flow the etchant liquid therethrough; and a second conduit affixed to a rear of the gas dispensation nozzle, attached to the clamp structure, and configured to flow the gas therethrough.

8. The method of claim 1, wherein the liquid dispensation nozzle is oriented such that a primary spray direction of the stream of the gas is parallel to a primary spray direction of the stream of etchant liquid from the liquid dispensation nozzle.

9. The method of claim 1, further comprising performing a pre-dispensation preparation process in which the stream of gas is turned on while the etchant liquid is not dispensed from the orifice of the liquid dispensation nozzle prior to performing the etchant liquid dispensation process.

10. The method of claim 9, further comprising performing a post-dispensation purge process in which the stream of gas is turned on while the etchant liquid is not dispensed from the orifice of the liquid dispensation nozzle after performing the etchant liquid dispensation process.

11. The method of claim 1, wherein the etchant liquid dispensation process comprises: a main dispensation step during which the stream of gas is turned on at a normal flow rate; a commencement step during which the stream of gas is turned on at a first enhanced rate that is higher than the normal flow rate; and a termination step during which the stream of gas is turned on at a second enhanced rate that is higher than the normal flow rate.

12. The method of claim 1, wherein: the device wafer comprises a semiconductor wafer including semiconductor devices and metal interconnect structures thereupon, and a layer of copper located over the semiconductor devices; and the stream of the etchant liquid etches the layer of copper located in the peripheral region of the top surface of the device wafer.

13. The method of claim 12, wherein the etchant liquid comprises a mixture of sulfuric acid and hydrogen peroxide, and the stream of the gas comprises air, nitrogen or argon.

14.-20. (canceled)

21. The method of claim 1, wherein: the liquid dispensation nozzle is tilted along a radial direction by a first angle relative to a horizontal plane so that the etchant liquid is dispensed outward along a radial direction; and the gas dispensation nozzle is tilted along the radial direction by a second angle relative to the horizontal plane so that the gas is dispensed outward along the radial direction.

22. The method of claim 21, wherein the second angle is not greater than the first angle.

23. The method of claim 21, wherein the first angle is in a range from 40 degrees to 55 degrees.

24. The method of claim 1, further comprising maintaining an alignment between the liquid dispensation nozzle and the gas dispensation nozzle during the etchant liquid dispensation process using a mechanical support structure.

25. The method of claim 1, further comprising rotating the device wafer at a variable speed during the etchant liquid dispensation process to control distribution of the etchant liquid on the top surface of the device wafer using a chuck rotation mechanism.

26. The method of claim 1, further comprising adjusting a flow rate of an etchant liquid and a flow rate of the gas based on real-time feedback from a sensor monitoring the edge bevel removal process.

27. The method of claim 1, further comprising monitoring a position of an edge of a dispensed portion of the etchant liquid on the top surface of the device wafer using a video camera.
